



PRODUCT DATA SHEET

INSULATING OIL – TECHNOFLUID TRANSFORMER OIL

Commercial / Pack Name: TRANSFORMER OIL

Product Description

- TECHNOFLUID Transformer Oil is a high-quality insulating mineral oil specially refined for use in electrical transformers and switchgear.
- Exhibits excellent dielectric strength, thermal stability, and oxidation resistance, ensuring reliable electrical insulation and efficient heat dissipation.
- Manufactured from carefully selected base stocks and processed to achieve high purity, low moisture content, and low sulphur levels.
- Designed to provide long service life while maintaining insulation performance under continuous electrical and thermal stress.
- Suitable for use in both new transformers and top-up of existing units (subject to compatibility checks).

Key Performance Benefits

- High dielectric strength ensures reliable electrical insulation.
- Excellent heat transfer properties aid in efficient cooling of transformer windings.
- Superior oxidation stability reduces sludge formation and extends oil life.
- Low moisture content minimizes risk of electrical breakdown.
- Non-corrosive and compatible with transformer construction materials.
- Helps maintain transformer efficiency and reduce maintenance frequency.

Applications

Recommended for use in:

- Power and distribution transformers
- Instrument transformers
- Oil-filled circuit breakers
- Switchgear and other oil-filled electrical equipment

Suitable for indoor and outdoor electrical installations.

Typical Uses

- Electrical insulation and cooling medium in high-voltage equipment.
- Systems requiring stable performance under continuous electrical load.
- New installations as well as maintenance and replacement applications.

Performance Standards Meets or exceeds:

- **IS 335 : 2018**
- **IEC 60296**
- **ASTM D3487** (*where applicable*)

Operating Characteristics

- Maintains stable dielectric and thermal properties over prolonged service life.
- Low gassing tendency under electrical stress.
- Compatible with commonly used transformer materials and sealing systems.

Typical Specifications

Characteristics	Test Method	Typical Figures
Appearance	Visual	Clear & Bright
Colour	ASTM D1500	≤ 0.5
Base Oil Type	—	Highly Refined Mineral
Density @ 15°C, g/ml	ASTM D4052	0.875
Kinematic Viscosity @ 40°C, cSt	ASTM D445	≤ 12.0
Kinematic Viscosity @ 100°C, cSt	ASTM D445	≤ 3.0
Viscosity Index	ASTM D2270	≥ 95
Flash Point (COC), °C	ASTM D92	≥ 140
Pour Point, °C	ASTM D97	≤ -6
Neutralization Value, mg KOH/g	ASTM D664	≤ 0.01
Moisture Content, ppm	ASTM D6304	≤ 30
Breakdown Voltage (BDV), Kv	IS 6792 / IEC 60156	≥ 60
Interfacial Tension @ 27°C, mN/m	ASTM D971	≥ 40
Copper Corrosion (3 hrs @ 100°C)	ASTM D130	1a
Oxidation Stability (Sludge), % wt	IEC 61125	≤ 0.1

Oxidation Stability (Acidity), mg KOH/g	IEC 61125	≤ 0.1
Sulphur Content (Corrosive)	ASTM D1275B	Non-Corrosive
Dielectric Dissipation Factor (Tan δ @ 90°C)	IEC 60247	≤ 0.005
Ash Content, % wt	ASTM D482	≤ 0.01
PCB Content	IEC / EPA	N